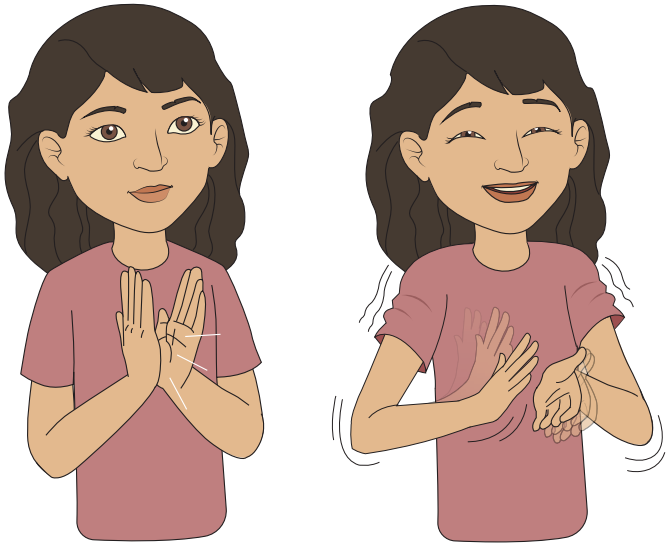
Collisions Transfer Energy

Chapter 5

Clap your hands. When one hand hits the other, your hands **collide**. Lift your foot, and then stomp down to make it collide with the floor. Roll two toy cars down ramps toward each other, and watch them bang into each other. That's a **collision**, too. A collision happens when two objects make contact.

In every collision, changes occur. You might see objects change the speed or direction of motion or even change shape because of a collision. You might hear a sound or see sparks from objects colliding. Or, you might feel an object colliding with you.



When you clap your hands, you can feel the results of a collision. The clap sounds and feels different if you clap slowly or quickly. What other changes occur during hand clapping?

Big Question

How is energy involved in collisions?

Vocabulary

collide, v. to come together with impact

collision, n. an instance of colliding

The word part *col-* in *collide* and *collision* means together or with.

Collisions Involve Changes in Motion

Moving objects have energy of motion—and in any collision, at least one object is moving. We know that energy is involved in collisions because collisions result in changes. Energy causes changes.

A bowling ball has motion energy as it rolls along the lane toward the pins. What happens when the bowling ball crashes into the pins? Can you predict how the pins will change?



The bowling ball's energy of motion is about to cause a change.

Before the collision, the pins are sitting still and not moving. The collision with the bowling ball changes that. The ball strikes the pins, and some of the ball's energy of motion transfers to the pins. This makes the pins gain motion energy and leaves the ball with less motion energy. The collision between ball and pins also produces a loud noise—*CRASH*!—a change in the form of energy from motion to sound.

What would happen if two bowling balls smashed into each other from opposite directions? The colliding balls would bounce off of each other, changing their motion. The speed and direction of each ball would change. The energy of motion would cause changes during this collision.

A Collision Can Change an Object's Shape

An object in motion has energy. In a collision, that energy can cause many kinds of changes, not just changes in motion.

Imagine you have a lump of clay rolled into a ball the same size as a baseball. You throw the clay hard against a brick wall. *SPLAT*! The collision doesn't set the wall in motion. The energy of motion causes a different change. First, the motion of the clay ball stops. Second, the shape of the clay becomes as flat as a pancake!

Think again about the two colliding bowling balls. They are hard enough that they can roll into each other, bounce apart, and have their shapes probably remain unaffected. But think about an egg that is dropped and collides with the floor. The shell shatters, and the liquid inside splatters into a mess. This is a change in shape.

Engineers design cars to protect passengers during a crash. Cars are designed to limit the damage to people that might occur due to a collision. The car in the picture is built so the front part changes shape more easily than the parts closest to the passengers. This design is on purpose. Less of the energy from the collision is transferred to the passengers, better protecting them from injury.



Cars are designed to protect passengers from collisions. The front of the car changes shape more easily than the inside where people sit.

Many Collisions Make Sounds

CLAP! Two hands collide. *CRASH*! Two cars collide. *CRACK*! A baseball and a bat collide. *SPLAT*! An egg and the floor collide. *BONG*! A mallet collides with a drum. Collisions often produce a **sound** as well as changes in motion and shape. When objects collide, some of the motion energy causes vibrations in the objects.

Vocabulary

sound, n. a form of energy that comes from a vibrating object

These vibrations also cause the air to vibrate. Those vibrations transfer energy to our ears. We sense this transfer of energy as sound. When two objects collide, the energy of motion causes changes that result in the energy of sound.



The collision between a mallet and a drum makes a big sound. The shape of the drum transfers much of the energy from the collision into the surrounding air, and we detect it as sound.

Think About Space

Collisions in space don't make sound. There is no air in the vacuum of space. So, there are no particles of matter to be affected by the vibrations of the collision.

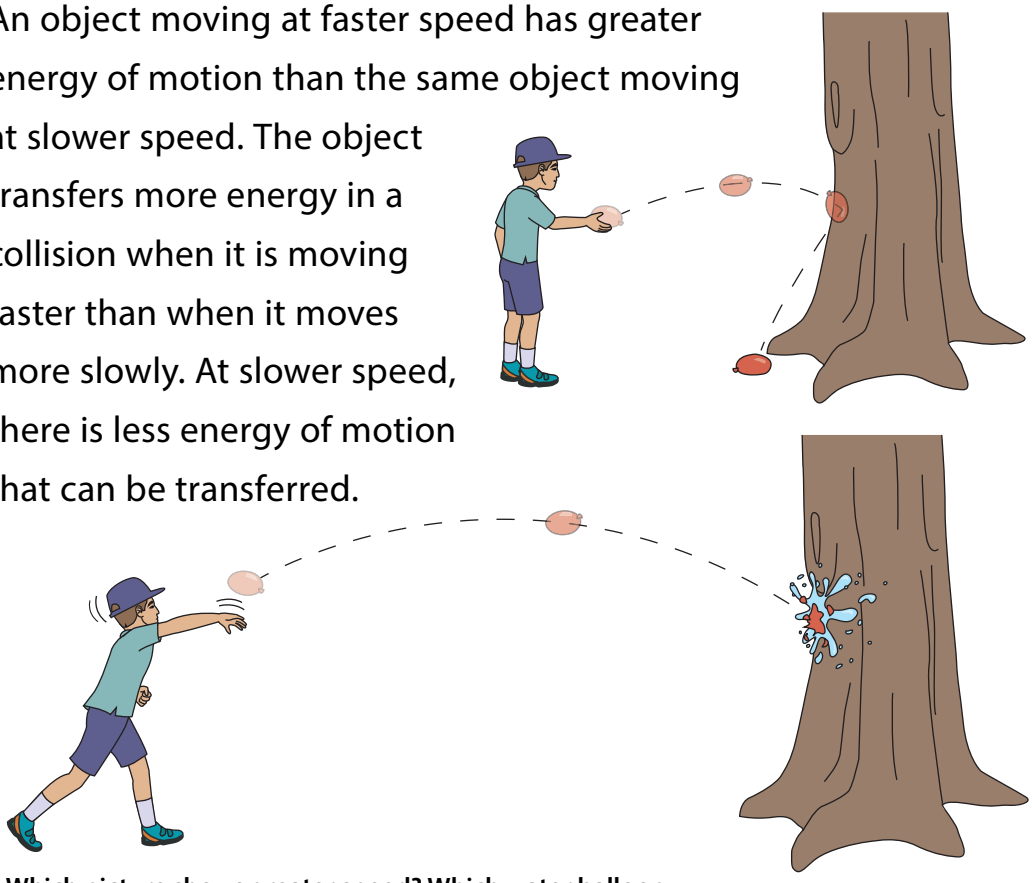
Without matter surrounding the colliding objects, sound energy cannot move away from the vibrating objects.

Speed Affects Collisions

Speed is how fast an object moves. In previous chapters, you learned that the speed of an object is related to its motion energy. It is not surprising that the speed of moving objects affects the kinds of changes that occur in collisions.

In the game of water balloon toss, the goal is to keep throwing water balloons back and forth without breaking them. The winners know they need to toss the balloons gently and catch them carefully. Throwing a water balloon faster produces a harder collision. Greater energy from the balloon's motion stretches the thin material until it tears and the water gushes out.

An object moving at faster speed has greater energy of motion than the same object moving at slower speed. The object transfers more energy in a collision when it is moving faster than when it moves more slowly. At slower speed, there is less energy of motion that can be transferred.

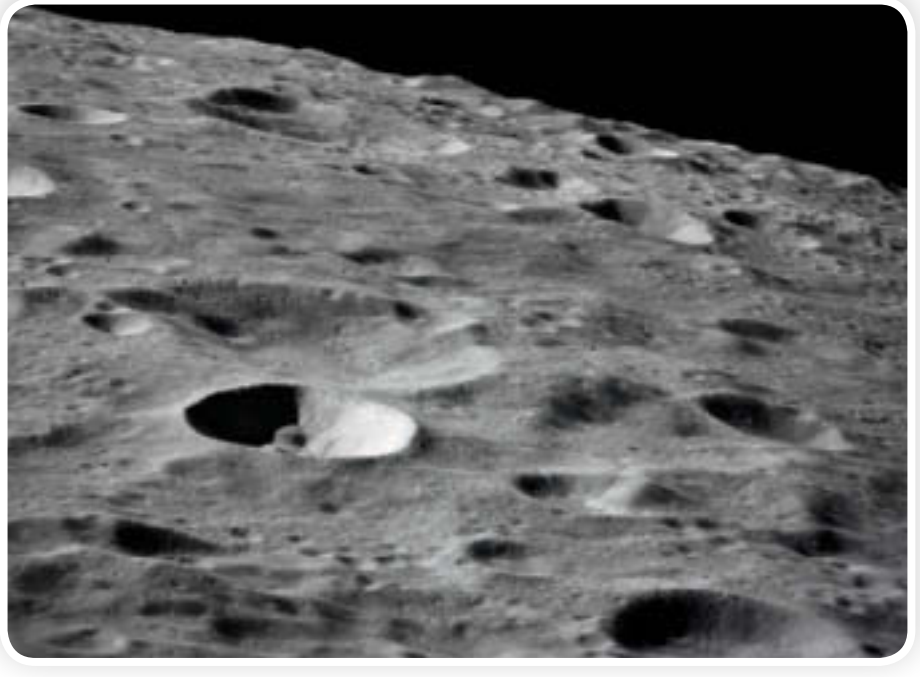


Which picture shows greater speed? Which water balloon had the greater energy of motion?

Weight Affects the Changes Caused by Collisions

Getting hit by a flying cotton ball is silly fun. Getting hit by a flying rock is a different story! Even if the cotton ball and the rock are the same size and moving at the same speed, the rock has greater energy of motion. Why? The rock weighs more. So, the rock has more energy of motion than the cotton ball moving at the same speed. Weight affects the energy of collisions.

Each moon crater offers evidence that motion energy during collisions causes change. Objects in space, such as meteoroids, have been crashing into the moon's surface for millions of years. The heavier these moving space rocks were, the more motion energy they brought to the collisions with the moon. Heavier meteoroids produced deeper craters than lighter meteoroids traveling at the same speed.



The speed and weight of meteoroids have resulted in craters of different sizes on the moon.